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Week 3 — Training Day 1: Operations Concept Development

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Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

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Classification: Internal

Training Day 1 — Operations Concept Development

Day at a glance

When	What	Output
09:00 – 10:30	Brief: ground-segment elements; ConOps recap; tools intro	Q&A captured
10:30 – 12:00	Mission Planning System (MPS) hands-on	First plan exported
13:00 – 15:00	MCS round-robin: each student rotates through six positions	Position tour completed
15:00 – 16:30	Pass-plan team exercise on the MPS	Team-written pass plan
16:30 – 17:00	Debrief & Day 2 brief; **expected reading flag**	Set up for Day 2

Setting: uOttawa Cyberrange, console room. Simulator runs the **Nominal Post-Commissioning** breakpoint (BP-1 in the breakpoint spec) — the spacecraft is healthy and doing routine ops, so the team can focus on tools rather than recovery.

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1. Brief — what we're flying

[Claude Code: expand to ~400 words. Cover EOSAT-1 mission: 6U CubeSat, ocean-current monitoring with multispectral imager (443, 560, 665, 865 nm), 450 km Sun-synchronous orbit at 98° inclination. Ground stations Iqaluit and Troll, both polar S-band. Reference `~/SpaceMissionSimulation/configs/eosat1/mission.yaml` and `orbit.yaml` for canonical values.]

1.1 The four ground-segment elements

[Claude Code: 200 words. Reference NASA Mission Operations Handbook structure: ground stations · baseband · MCS · flight dynamics. EOSAT-1 specifics: Iqaluit (63.7° N), Troll (72° S), S-band, 5° minimum elevation.]

1.2 Pass geometry — the polar S-band picture

[Claude Code: 200 words and a diagram. Show why polar SSO with polar ground stations gives many short passes per day. Include a Python-generated diagram (use `outputs/sample_figures/uottawa_brand.py` style) showing one orbit with ground tracks, Iqaluit + Troll visibility cones, and AOS/TCA/LOS markers.]

1.3 Procedure structure

[Claude Code: 150 words. Walk through one procedure from the simulator (recommend `procedures/leop/first_acquisition.md` since it's the simplest LEOP one). Highlight: preconditions block, ownership column, verification, exit criteria. Tell students they will read procedures, not write them.]

1.4 PUS service primer (light)

[Claude Code: 150 words. Introduce only services 1, 3, 5, 8, 9, 11, 17 by name and one-line meaning. Source: `mcs/pus_services.yaml`. Emphasise that students use the MCS UI; PUS knowledge is for context only.]

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2. EOSAT-1 mission summary (expected reading before Day 1)

> **Read before Day 1.** Approximate reading time 25 minutes.

[Claude Code: produce a 600 – 900 word readable summary covering: spacecraft platform (6U bus, multispectral imager, S-band), orbit choice rationale (Sun-sync 450 km, why), payload science driver (ocean-current monitoring, why those 4 bands), ground segment, ops concept summary. Include diagrams: orbit + ground tracks + ground station visibility (Python figure); spacecraft block diagram with six MCS-position colour-coding; pass-cadence chart.]

Key references

- EOSAT-1 mission config: `configs/eosat1/mission.yaml`
- ECSS-E-ST-70-11C — Space segment operability — https://ecss.nl/standards/active-standards/ecss-e-st-70-11c-space-segment-operability/
- NASA *Space Flight Program and Project Management Requirements* (NPR 7120.5F) — https://nodis3.gsfc.nasa.gov/displayDir.cfm?Internal_ID=N_PR_7120_005F_
- CCSDS Mission Operations Services Concept (CCSDS 520.0-G) — https://public.ccsds.org/Pubs/520x0g4.pdf

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3. Mission Planning System hands-on

Tool: the MPS web UI (port 9091 by default — the planner's own `--port`; the simulator uses 8080 and the MCS 9090).

[Claude Code: produce a 1.5-page exercise. Use planning configs (`planning/activity_types.yaml` , `planning/imaging_targets.yaml` , `planning/ground_stations.yaml`) as reference. Steps: 1. Open MPS, view ground-station schedule for next 24 h 2. Pick three ocean-current imaging targets from the config 3. Drop one imaging activity per target into the next available viable window 4. Run conflict check (pointing slew time, power, thermal) 5. Resolve conflicts, export plan 6. Open the resulting plan in the MCS procedure panel.]

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4. MCS round-robin

Six MCS positions, six 15-minute stations. Every student visits all six during the afternoon. At each station, the seat-holder reads the position's role analysis card aloud, then completes one position- specific task while watching live telemetry.

Position	Role analysis source	15-min task
Flight Director	`mcs/role_analysis/flight_director_role.md`	Run a GO/NO-GO round on the live dashboard
Power & Thermal	`mcs/role_analysis/eps_tcs_role.md`	Identify the next eclipse and predict battery SoC at exit
Flight Dynamics (AOCS)	`mcs/role_analysis/aocs_role.md`	Compute AOS, TCA, LOS for the next Iqaluit pass
TT&C		

Position	Role analysis source	15-min task
	`mcs/role_analysis/ttc_role.md`	Configure low-rate beacon → high-rate downlink at AOS
Payload Operations	`mcs/role_analysis/payload_ops_role.md`	Schedule a single imaging session via MPS
FDIR / Systems	`mcs/role_analysis/fdir_systems_role.md`	Walk through the FDIR panel and explain one rule

[Claude Code: convert each row into a one-page station card and include in the student bundle.]

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5. Pass-plan team exercise

Team produces a written 60–90-min pass plan for an upcoming Iqaluit pass. Plan must include:

- Pass timing (AOS – TCA – LOS, with margins)
- Procedures to run (using the curated procedure quick-reference)
- Imaging window, target ID, predicted ground swath
- Downlink config (data rate, encoding)
- Anomaly contingency: name **two** plausible failures and the procedure ID to run for each

Plans are read aloud at 16:30 and critiqued by another team using the team-axis peer-review worksheet.

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6. Debrief & expected reading for Day 2

> **Expected reading before Day 2.** Read the *Student MCS & MPS Reference* (`student_tool_reference.md`) — sections 1 to 4. Approximate reading time 45 minutes. Bring questions for Day 2 09:00.

Debrief questions:

1. Which MCS position felt most natural to you? Why?
2. What surprised you about a ground-station pass timing-wise?
3. What's one thing you'd want documented better before Day 2?

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7. Tools, files & references

Tools used

- Mission Control System (MCS) — webapp, port 9090
- Mission Planning System (MPS) — webapp, port 9091 (its own server)
- Voice loop — Cyberrange softphone

Simulator files referenced

- `configs/eosat1/mission.yaml`

- `configs/eosat1/orbit.yaml`
- `configs/eosat1/mcs/positions.yaml`
- `configs/eosat1/mcs/role_analysis/*.md`
- `configs/eosat1/planning/*.yaml`
- `configs/eosat1/procedures/procedure_index.yaml`

Open references

- ECSS-E-ST-70C — Ground systems and operations — https://ecss.nl/standards/active-standards/ecss-e-st-70c-ground-systems-and-operations/
- ECSS-E-ST-70-11C — Space segment operability — https://ecss.nl/standards/active-standards/ecss-e-st-70-11c-space-segment-operability/
- CCSDS standards — https://public.ccsds.org/Publications/default.aspx
- NASA Space Flight Program and Project Management Requirements (NPR 7120.5F) — https://nodis3.gsfc.nasa.gov/displayDir.cfm?Internal_ID=N_PR_7120_005F_